Gregory Polyakov

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Summary

Third-year PhD student at the University of Tübingen and IMPRS-IS working on mechanistic interpretability of language models for NLP and IR. I study how models implement complex behaviors such as hallucination, in-context learning, and relevance estimation, and turn these findings into methods for controlling and improving them. First author of four papers, including at EMNLP and SIGIR, with three years of industrial R&D at Huawei Noah's Ark Lab on production search and voice-assistant systems.

Education

University of Tübingen & Max Planck Research School — *Tübingen, Germany*
PhD *in Computer Science* — *Oct 2024 - Present*

- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research focus: Mechanistic interpretability of language models for NLP and IR tasks.

Moscow Institute of Physics and Technology — *Moscow, Russia*
MSc *in Applied Mathematics and Computer Science* — *Sep 2021 - Jul 2024*

- Research topics: Sparse Retrieval, Multimodal Retrieval, Tool Calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

BSc *in Applied Mathematics and Computer Science* — *Sep 2017 - Jul 2021*

- Thesis: "Usage of MCMC Methods in Deep Learning".

Work experience

Huawei Noah's Ark Lab — *Moscow, Russia*
NLP Research Engineer, full-time — *Nov 2022 - Sep 2024*

- Designed and co-implemented the end-to-end NLU pipeline for a production in-car voice assistant in Russian.
- Developed a fine-tuning method that teaches LLMs to detect and recover from failed API calls (ACL 2025 REALM Workshop).
- Co-developed a post-processing method for text-video retrieval (SIGIR 2023).

NLP Research Intern, full-time — *Oct 2021 - Nov 2022*

- Co-developed the reranking model for a large-scale Russian-language product search engine: improved ranking quality by 5% relative NDCG@10 and cut inference latency 1.5x by adapting ColBERTer and TinyBERT-style distillation.

Publications

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance — *SIGIR 2026*
G. Polyakov*, H. Scells, C. Eickhoff*

- Introduced metrics to formally quantify wacky weights (semantically unrelated expansion terms) in SPLADE sparse retrieval models.
- Showed that wacky weights mainly drive in-domain effectiveness, and measured which SPLADE settings promote more wacky weights.

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models *EMNLP 2025*

***G. Polyakov**, C. Hepting, C. Eickhoff, A. Bahrainian*

- Identified the internal mechanisms LLMs use for in-context learning on arithmetic tasks.
- Used these findings to design a steering method that improves zero-shot performance, reducing the need for few-shot examples.

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval *SIGIR 2025*

***G. Polyakov**, C. Chen, C. Eickhoff*

- Proposed a noise-mitigating evaluation metric for activation patching in neural rankers that filters out spurious low-confidence signals.
- Systematically analyzed diagnostic datasets, identifying key selection criteria for queries and documents to ensure robust interpretability analysis.

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data *ACL 2025 REALM Workshop*

***G. Polyakov**, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya*

- Proposed a fine-tuning strategy that enables LLMs to self-correct API calls by learning from runtime execution feedback.
- Co-developed evaluation datasets designed to assess tool-augmented LLMs in more realistic scenarios.

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval *SIGIR 2023*

*K. Yakovlev, **G. Polyakov**, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya*

- Co-developed a post-processing method based on Sinkhorn transformations that improves video retrieval quality.
- Introduced a realistic evaluation setting to compare post-processing methods in scenarios closer to real-world applications.

Funding & Scholarships

Presidential Grant Scholarship *Feb 2018 - Aug 2021*

One of 1,200 recipients nationwide; four-year stipend awarded for academic merit and competition results.

Teaching Experience

University of Tübingen

Teaching Assistant, Introduction to AI *Winter Semester 2026/27*

- Conducted hands-on tutorials and evaluated student assignments.

Teaching Assistant, Modern Search Engines *Summer Semesters 2025, 2026*

- Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant, Introduction to Python Programming *Winter Semesters 2024/25, 2025/26*

- Conducted hands-on tutorials and evaluated student assignments.

Skills

Programming: Python, C/C++, Bash.
ML & NLP: PyTorch, Hugging Face Transformers, PEFT, Scikit-learn, CatBoost.
Interpretability: TransformerLens, causal mediation analysis, steering, probing, sparse autoencoders, cross-layer transcoders.
IR: Pyserini, FAISS, sparse and dense retrieval, neural reranking.
Infrastructure: SLURM, multi-GPU training, Git.
Languages: Russian (native), English (C1, IELTS Band Score 7).